

Performance Augmentation of GaInP/GaAs Multi-Junction Solar Cell: A PINN Driven Approach to Shallow Doped Layer Optimization

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Abstract: This work introduces an innovative approach to modeling and predicting the performance matrices of solar cells by integrating physics based TCAD simulation with advanced machine learning techniques, specifically Physics-Informed Neural Networks (PINNs). By leveraging fundamental physical principles, this method provides deeper insights into key interactions such as electrostatic potential distribution, ultimately enabling enhanced solar cell performance. The proposed model incorporates partial differential equations to simulate the device's physical behavior while utilizing PINNs to predict efficiency calculation with high accuracy. In this study, we have investigated the performance of a shallow doped GaAs layer alongside the intrinsic layer in terms of doping concentration and thickness in the lattice-matched GaInP/GaAs multi-junction solar cell with an AlInP cap layer and an MgF₂/TiO₂ anti-reflection coating. The predictive model is validated using Sentaurus TCAD simulations, providing insights into the impact of key structural parameters on photovoltaic performance. We systematically analyze the influence of intrinsic layer thickness, shallow layer doping and thickness on critical performance metrics, including open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and power conversion efficiency (PCE). Our optimized solar cell design, featuring a 0.06- μ m-thick intrinsic layer, 1.4 μ m-thick shallow doped layer with 1017 cm⁻³ doping concentration with a Ga_{0.5}In_{0.5}P/GaAs active region, achieves a PCE of 33.14% representing a 68% improvement over our previously reported designs. These findings highlight the potential of structural and material optimizations in enhancing III-V solar cell performance, paving the way for next-generation machine learning based high-efficiency photovoltaic applications.

Keyword- Multi-junction, Shallow layer, PINN and PCE

I. Introduction

The global transition toward sustainable and clean energy sources has intensified research in high-efficiency photovoltaic (PV) technologies. Among the various available photovoltaic technologies, III-V semiconductor-based multijunction solar cells have garnered considerable attention due to their superior opto-electronic properties, wide bandgap tunability, excellent lattice-matching compatibility, high absorption coefficients, and long carrier lifetimes [1-8]. In particular, GaInP/GaAs-based pin solar cells have emerged as a promising candidate for high-performance

photovoltaic devices, enabling efficient utilization of the solar spectrum through careful bandgap engineering [9-15]. These devices exhibit enhanced carrier collection efficiency and reduced recombination losses due to well-structured hetero-interfaces. Extensive efforts have been dedicated to optimizing the device structure of GaInP/GaAs PIN solar cells. Prior studies have demonstrated the significant influence of intrinsic layer thickness, cap layer composition, and transparent conductive contacts on the overall device performance [16-26]. For example, Li et al. proposed a GaAs/InGaAs quantum well pin solar cell grown on a GaAs substrate to improve carrier collection, although this design encountered issues related to misfit strain and reduced open-circuit voltage (Voc) [1]. Hasem et al. [2] fabricated GaInP-based solar cells using MOCVD techniques, successfully addressing current matching challenges but facing limitations due to quantum confinement effects. Recent efforts by Nath and Biswas [3] introduced a lattice-matched p-GaInP/i-GaAs/n-GaInP heterojunction solar cell featuring an AlInP cap layer and an ITO contact. Their work demonstrated that optimizing the intrinsic layer thickness to 2.1 μm and the cap layer thickness to 156 nm could enhance the power conversion efficiency (PCE) to 19.89%, representing a 67% improvement over conventional designs. While device optimization through TCAD simulations and analytical modeling has yielded significant advancements, these methods are computationally intensive and often require iterative tuning of numerous parameters. Additionally, the predictive capability of conventional models is constrained by their reliance on predefined approximations and assumptions. In recent years, machine learning (ML) and deep learning techniques have been applied to accelerate modeling and prediction tasks in photovoltaic research. However, traditional ML approaches require large experimental datasets and lack the ability to inherently incorporate governing physical laws, limiting their generalization capabilities across diverse design spaces. To address these challenges, Physics-Informed Neural Networks (PINNs) have emerged as a powerful modeling tool. First introduced by Raissi et al. [4], PINNs integrate partial differential equations (PDEs), such as the Poisson equation and carrier continuity equations, directly into the neural network framework. This approach allows PINNs to learn the underlying physics of complex systems with minimal experimental data while maintaining physical consistency and accuracy. Recent studies by Taboria Baelongandi et al. [5] have demonstrated the applicability of PINNs in predicting electrostatic potential distributions in solar cells under thermodynamic equilibrium, laying the groundwork for more comprehensive non-equilibrium modeling in the future. Despite the demonstrated potential of PINNs, their application to complex, multi-layered III-V solar cell architectures remains underexplored. In particular, integrating PINNs for the modeling, optimization, and prediction of GaInP/GaAs modified pin solar cell performance presents a significant opportunity to overcome computational barriers and accelerate device design steps.

In this paper, we propose a novel approach for photovoltaic performance enhancement of GaInP/GaAs modified pin solar cells using Physics-Informed Neural Networks. In this work, we have investigated three device optimizations to obtain optimized power conversion efficiency i.e. i. shallow layer thickness variation, ii. Variations of doping density of shallow layer and iii. Intrinsic layer

width variations. Our work has been developed on prior structural optimization efforts and aims to bridge the gap between conventional physics-based simulation and data-driven prediction. By incorporating physical models and boundary conditions into neural network training, we demonstrate that PINNs can effectively predict device performance parameters such as open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and power conversion efficiency (PCE) while providing deeper insights into carrier transport dynamics and electric field distribution. This study not only validates the accuracy and efficiency of PINNs in modeling high-efficiency III-V solar cells but also establishes a foundation for future research into AI-driven photovoltaic device optimization.

II. Device Architecture

In this work, we present a modified monolithic p⁺ Ga_{0.5}In_{0.5}P/p-GaAs/i-GaAs/n-GaAs/n⁺ Ga_{0.5}In_{0.5}P PIN solar cell structure, illustrated in Fig. 1. The fabrication of this advanced design can be achieved through the metal-organic chemical vapor deposition (MOCVD) technique, as successfully demonstrated by Yao et al. [12]. The active region is meticulously engineered using epitaxial growth, consisting of sequential layers of p⁺ Ga_{0.5}In_{0.5}P (5 nm), p-GaAs (140 nm), i-GaAs (6 nm), n-GaAs (140 nm), n⁺ Ga_{0.5}In_{0.5}P (5 nm), and a 600 nm n-GaAs substrate.

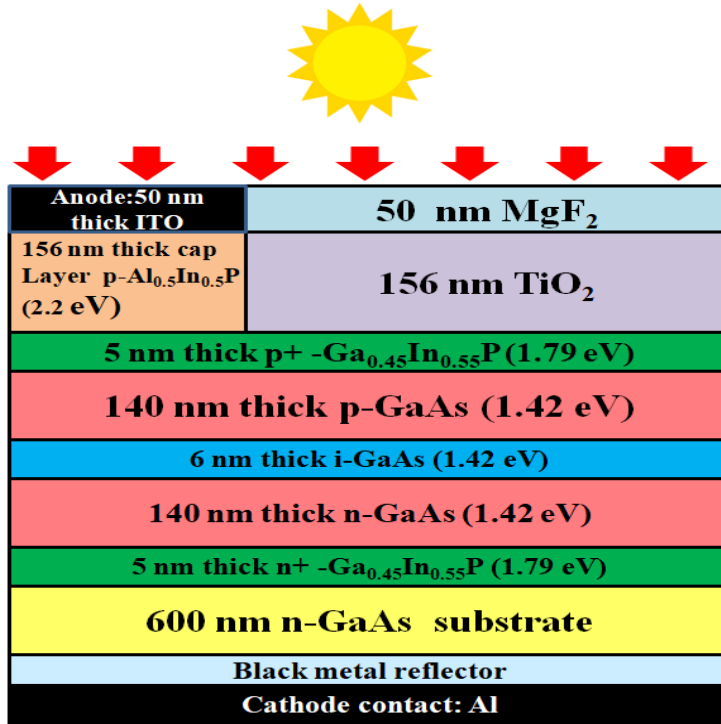


Fig. 1 Schematic device architecture of the proposed GaInP/GaAs M solar cell

GaAs (140 nm), and n^+ $\text{Ga}_{0.5}\text{In}_{0.5}\text{P}$ (5 nm). These layers are carefully doped with concentrations of 10^{19} cm^{-3} for the p^+ and n^+ regions, and 10^{18} cm^{-3} for the p and n regions, all grown on an arsenic-doped n -GaAs (001) substrate with a doping concentration of 10^{18} cm^{-3} and a 6° offcut toward the nearest (111) A plane. Above the active region, a 156 nm thick p^+ - $\text{Al}_{0.4}\text{In}_{0.6}\text{P}$ cap layer is introduced with an optimal doping density of 10^{19} cm^{-3} . To form a ridge structure for accommodating anti-reflection coatings, conventional photolithography and dry etching techniques are applied. The anti-reflective coating, consisting of 100 nm MgF_2 and 56 nm TiF_2 layers, is deposited via sputtering. Finally, an ITO (Indium Tin Oxide) layer, a conductive n -type degenerate semiconductor, which is sputtered on top of the cap layer to serve as a back reflector and cathode contact. Metal contact layers are then defined using precise electron beam deposition, completing the sophisticated structure of the proposed solar cell.

III. Methodology

In this work, we develop a hybrid modeling framework that combines physics-based device architecture optimization with Physics-Informed Neural Networks (PINNs) to predict and enhance the photovoltaic performance of a GaInP/GaAs modified PIN solar cell. The methodology comprises three main stages: (1) Simulation framework and model calibration using TCAD, (2) formulation of the governing physics equations, and (3) development and training of the PINN model for performance prediction and optimization.

1. *Simulation framework and model calibration using TCAD:*

Using SENTAUROS TCAD, we perform a thorough numerical study for multi-junction (MJ) solar cells using the Transfer Matrix method as an optical solver under a 1 sun condition of the AM1.5G spectrum. The simulator firstly captures the photo-absorption and photo-carrier generation within the solar cell invoking Quantum field model. Using the "Wavelength Dep" model in the simulation, the optical performance matrices such as optical intensity, absorbed photon density, optical generation, spectral response, and quantum efficiencies have been extracted in order to incorporate the wavelength dependent absorbance within the cell. The effect of non-radiative recombination centers within the cell, which directly control the carrier collecting efficiency, has been captured by the SRH recombination model. Next, to calculate the electrical characteristics, Jain Roulston band bending model and band gap narrowing model have been used to capture the electrical performance parameters and to investigate the impact of heavy doping on band gap narrowing. To estimate the carrier distribution as a function of energy, Fermi-Dirac statistics is used. Our numerical simulation involves coupled solution of Poisson and Schrodinger equation for charge carriers. Finally, the transport of carriers is finally modelled using the Drift-Diffusion transport model. This has been accomplished by using the doping dependent mobility model, effective masses of electrons and holes, densities of states of electrons and holes in the conduction band and valence band,

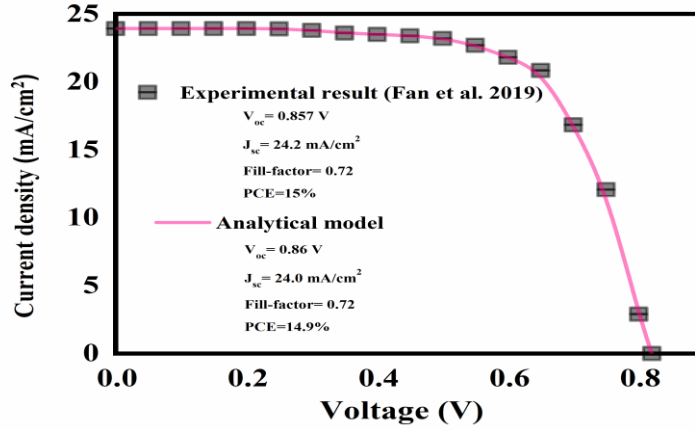


Fig. 2 Calibration of the simulation deck with experimentally published data

respectively, and saturation velocities of electrons and holes. Fig. 2 compares the simulated current-density (J) - voltage (V) characteristics for the selected conventional structure of the solar cell with the experimentally obtained data in [4]. Our simulated data and reported data manifest a very good agreement as illustrated in Fig. 2. This validates the authenticity of our chosen models and parameters used in the simulation.

2. Formulation of the governing physics equations:

I. Optical response formulation:

a. Incident Light and Interface Reflection

At each interface between two materials with refractive indices (n_1) and (n_2), the reflectance (R) and transmittance (T) at normal incidence are given by the Fresnel equations:

$$R(\lambda) = |(n_1(\lambda) - n_2(\lambda)) / (n_1(\lambda) + n_2(\lambda))|^2 \quad (1)$$

$$T(\lambda) = 1 - R(\lambda) \quad (2)$$

For the current work, these can be extended using the transfer matrix method.

b. Transfer Matrix Method for Multi-Junction Solar Cells

For a multi-junction solar cell composed of several layers with different refractive indices and thicknesses, the optical field distribution and absorption can be accurately modeled using the transfer matrix method (TMM). Each interface between layers and propagation through each layer are considered to account for interference, reflection, and transmission effects.

c. Transfer Matrix Components

At normal incidence, for each layer with refractive index (n_j), thickness (d_j), and wave vector ($k_j = 2\pi n_j/\lambda$), the propagation matrix (P_j) and interface matrix ($I_{j,j+1}$) are defined as:

Propagation Matrix:

$$P_j = \begin{bmatrix} \exp(ik_j d_j) & 0 \\ 0 & \exp(-ik_j d_j) \end{bmatrix} \quad (3)$$

Interface Matrix:

$$I_{j,j+1} = \frac{1}{t_{j,j+1}} \begin{bmatrix} 1 & r_{j,j+1} \\ r_{j,j+1} & 1 \end{bmatrix} \quad (4)$$

Where: $r_{j,j+1} = (n_j - n_{j+1}) / (n_j + n_{j+1})$

$t_{j,j+1} = 2n_j / (n_j + n_{j+1})$

d. Total Transfer Matrix

The total transfer matrix (M) for a stack of layers is:

$$M = \prod_{j=1}^N I_{j,j+1} P_j \quad (5)$$

From the total transfer matrix, the reflectance ($R(\lambda)$) and transmittance ($T(\lambda)$) at the front surface are calculated as:

$$R(\lambda) = |M_{21}/M_{11}|^2 \quad (6)$$

$$T(\lambda) = |1/M_{11}|^2 \times (n_{N+1}/n_1) \quad (7)$$

e. Generation Rate Calculation

The intensity of the optical field inside each layer $E_j(x, \lambda)$ is obtained from TMM results, and the generation rate is:

$$G_j(x, \lambda) = \alpha_j(\lambda) \times |E_j(x, \lambda)|^2 \varphi_{in}(\lambda) \quad (8)$$

f. Total Absorbed Photon Flux

Integrate over the spectrum to find the total generation profile:

$$G(x) = \int G_j(x, \lambda) d\lambda \quad (9)$$

g. Internal Electric Field Intensity (Thin Film Interference)

The intensity of light inside the active layers is influenced by multiple reflections and interference. The field distribution at position (x) can be expressed as:

$$I(x, \lambda) = I_0(\lambda)(1 - R_{front}(\lambda))\exp(-\alpha(\lambda) x) \quad (10)$$

2. Electrical Response Formulation

The electrical behavior is governed by Poisson's equation and the continuity equations for electrons and holes.

a. Poisson's Equation:

$$d^2\psi/dx^2 = -q/\epsilon \cdot (p - n + N_D^+ - N_A^-) \quad (11)$$

b. Electron Continuity Equation:

$$dJ_n/dx = q (G(x) - R_n(x)) \quad (12)$$

$$J_n = q\mu_n n \, d\psi/dx + q \cdot D_n \, dn/dx \quad (13)$$

c. Hole Continuity Equation:

$$dJ_p/dx = -q(G(x) - R_p(x)) \quad (14)$$

$$J_p = q\mu_p p \, d\psi/dx - q \cdot D_p \, dp/dx \quad (15)$$

d. Recombination Model (SRH recombination):

$$R_{SRH} = (np - n_i^2) / [\tau_p (n + n_i) + \tau_n (p + p_i)] \quad (16)$$

These PDEs are solved numerically and also form the core constraints embedded into the loss function of the PINN.

The PINN model is constructed using a fully connected deep neural network architecture with multiple hidden layers and hyperbolic tangent (tanh) activation functions. The input to the network is the spatial coordinate across the device thickness, and the outputs are the predicted electrostatic potential, electron density, and hole density. The loss function comprises three components: PDE residual loss enforcing that the PINN solutions satisfy the Poisson and continuity equations, boundary condition loss enforcing Dirichlet conditions at the device contacts and data loss comparing the PINN outputs with TCAD-simulated reference data for validation. The training process utilizes the L-BFGS optimizer combined with adaptive learning rate schedules. Collocation points are selected using Latin Hypercube Sampling (LHS) to ensure uniform spatial sampling across the device domain.

Results & Discussion:

In this section, we present the comparative analysis of simulated and PINN-predicted photovoltaic performance for three key structural parameters of the

GaInP/GaAs modified pin solar cell: (1) shallow doped region thickness, (2) shallow doped region doping concentration, and (3) intrinsic region thickness on the dint of four figures of merit i.e. V_{oc} , J_{sc} , FF, and PCE.

1. Optical Response:

The optical performance of solar cells is characterized by variations in internal quantum efficiency (IQE), external quantum efficiency (EQE), reflectance, and absorbance spectra across different wavelengths. In this study, two layers of GaAs with doping concentration of 10^{18} cm^{-3} are stacked sequentially (as shown in Fig. 1) on either side of intrinsic layer along with an AlInP cap layer. These layers are engineered to tailor their energy band gaps, enabling effective absorption of solar radiation across multiple regions of the solar spectrum. Fig.3 illustrates the calibrated IQE and EQE responses between the proposed device with conventional device, which are critical indicators of a solar cell's optical behavior. The IQE represents the ratio of generated electron-hole pairs (EHPs) to the number of absorbed photons within the active material, while the EQE measures the proportion of incident photons that are successfully converted into photocurrent. As shown in Fig.3, the IQE begins to rise from 359 nm and reaches a maximum value of 0.98 before declining sharply beyond 755 nm.

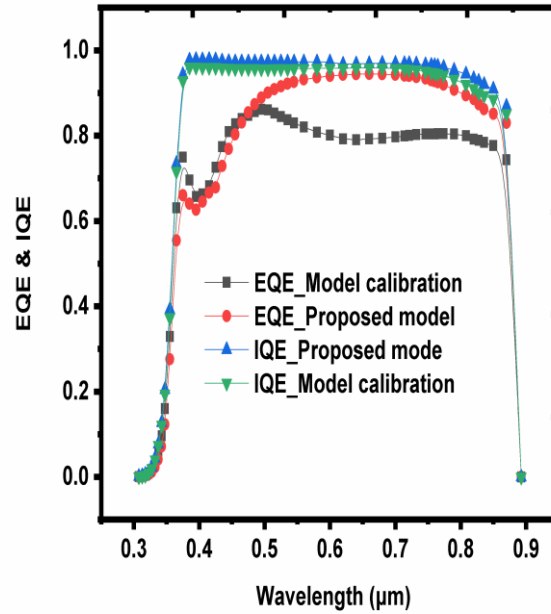


Fig. 3 Variation of IQE & EQE with optical wavelength

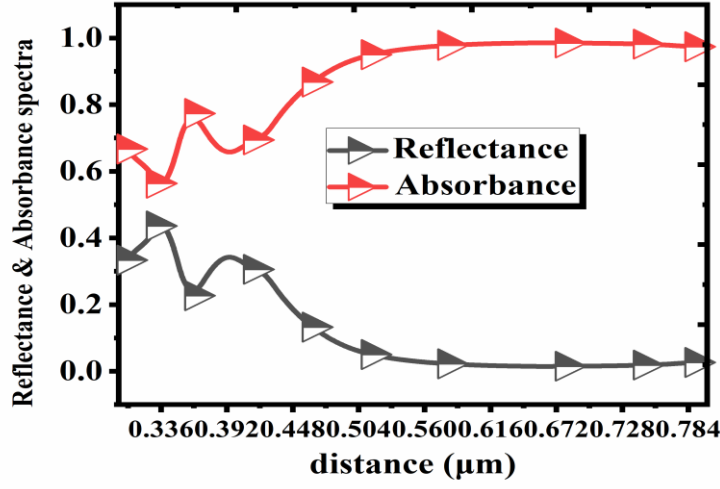


Fig. 4 variation of absorbance & reflectance spectra with wavelength

Similarly, the EQE starts increasing from 417 nm, peaks at 0.88, and both IQE and EQE exhibit a clear cut-off at 755 nm. This cut-off point (875 nm) corresponds to the bandgap of the GaAs layer used as the active material. The inclusion of the AlInP cap layer also significantly reduces front-surface recombination losses. The inclusion of the lightly doped GaAs region significantly boosts the photo-absorption inside the active region significantly by tailoring the bandgap. Fig. 4 depicts the variation of spectral response in terms of absorbance and reflectance with the optical wavelength and the figure agrees the same behavior as the previous figure portrays. The 2D contour plot of SRH and radiative recombination with optical generation is depicted in Fig. 5(a) & (b) which gives an in-depth insight of EQE and IQE.

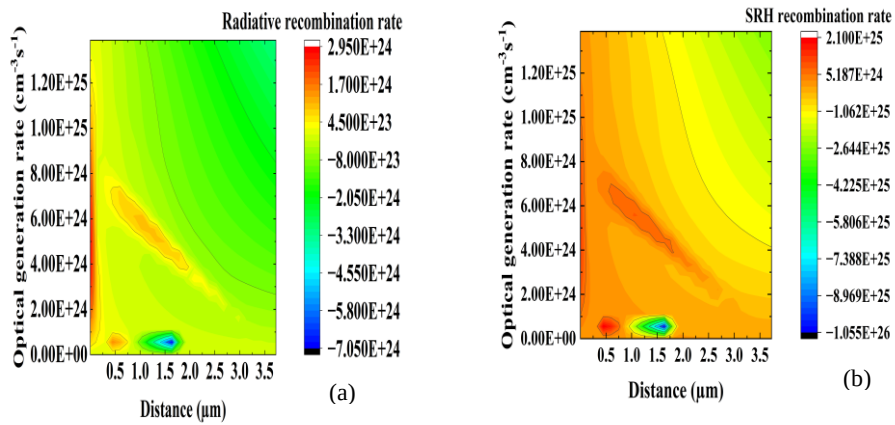


Fig. 5 2D Contour plot of optical generation with distance for (a) radiative & (b) SRH recombination rate

2. Effect of Shallow Doped Region Thickness:

The shallow doped region thickness directly influences the carrier transport and recombination near the surface. Fig.6 illustrates the dependence of performance matrices V_{oc} , J_{sc} , FF, and PCE on the lightly doped region thickness and the comparison between the simulated results and PINN induced predicted data have been presented for each value of shallow layer thicknesses ranging from 0.8 μm to 1.5 μm . Fig. 6(a) shows that V_{oc} remains consistently high around 1.276 V for thicknesses up to 1.2 μm , then slightly decreases to 1.234 V at 1.4 μm , indicating the optimal thickness range is around 1.0–1.2 μm for maintaining high voltage. The shallow doped region determines carrier transport from the active region to the external contacts controlled by the electric field. V_{oc} remains stable for thicknesses up to 1.2 μm and then reduces slightly. Thicker shallow layers beyond critical value reduces V_{oc} because of the increased recombination events. From Fig.6(b) it is observed that, J_{sc} increases from 29,538 mA/cm^2 at 0.8 μm to 30,276 mA/cm^2 at 1.4 μm under 1000 condition, indicating better carrier collection with increasing shallow layer thickness. J_{sc} peaks at around 1.4 μm , indicating that moderately thicker shallow layers enhance light capture and transport without excessive recombination but increase in layer thickness beyond 1.4 μm captures the effect of low absorption inside the active region as the optical path length is being hindered with the thicker layer. Fill factor (FF) also shows improvement and follows the similar pattern as V_{oc} does. The peak of FF reaches to 0.894 at 1.4 μm , indicating reduced series resistance and improved quality as depicted in Fig. Fig. 6(c). PCE rises steadily and reaches the peak value at 33.24% for 1.4 μm thickness (Fig. 6(d)). PCE follows the trend of J_{sc} and FF, peaking around 1.4 μm , beyond which excessive thickness reduces efficiency. A strong concordance between the simulated results and predicted values has been shown in the given figure. This analysis confirms that a shallow doped region thickness between 1.2 and 1.4 μm is optimal, balancing increased carrier generation with minimal recombination losses, well-predicted by the PINN model.

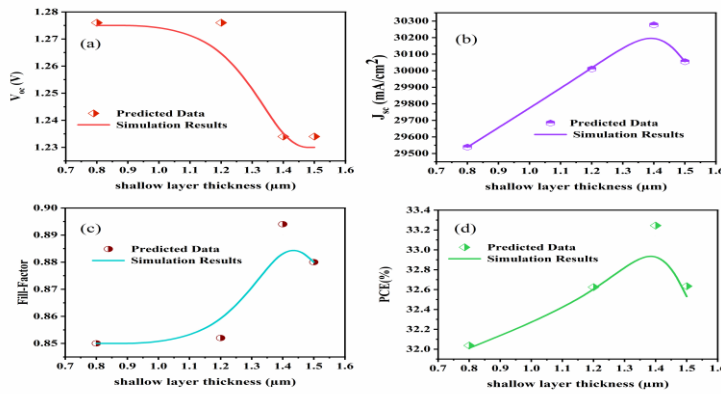


Fig. 6 Variation of FOMs (a) V_{oc} , (b) J_{sc} , (c) FF & (d) PCE for shallow region thickness

3. Variation of Shallow Doped Region Doping Concentration

Doping concentration in the shallow doped layer impacts both carrier mobility and recombination rates. The comparative study between the simulated results and PINN induced predicted data has been presented in Fig. 7 with doping concentrations ranging from 10^{16} to 10^{19}cm^{-3} , reveals that V_{oc} increases significantly at higher doping levels as higher doping strengthens the built-in potential and reduces recombination near the junction, peaking at 1.28 V at 10^{18}cm^{-3} , after that V_{oc} drops at 10^{19}cm^{-3} due to increased SRH recombination (Fig. 7(a)). From Fig. 7(b), J_{sc} follows a similar trend, increasing from 30,066 mA/cm^2 at 10^{16}cm^{-3} to 30,276 mA/cm^2 at 10^{18}cm^{-3} , then dropping to 29,066 mA/cm^2 at 10^{19}cm^{-3} . At very high doping, excessive impurity scattering and SRH recombination are two dominant factors which reduce carrier mobility and limit J_{sc} . Fill factor remains high and stable at 0.87–0.89, with its peak corresponding to the highest efficiency at 10^{18}cm^{-3} as shown in Fig. 7(c). The PCE reaches its maximum at 33.15% for 10^{18}cm^{-3} , followed by a slight decline at 10^{19}cm^{-3} due to carrier recombination losses (Fig. 7(d)). These results emphasize that a doping concentration of 10^{18}cm^{-3} provides the best trade-off between high voltage, high photocurrent, and minimal recombination. The PINN predictions match simulation results extremely closely, with variations well below 0.05%.

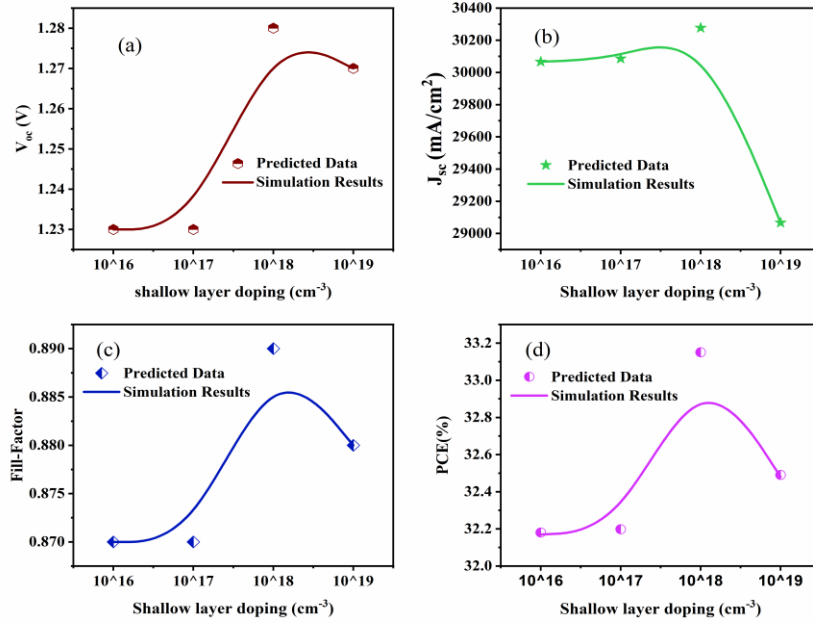


Fig. 7 Variation of FOMs (a) V_{oc} , (b) J_{sc} , (c) FF & (d) PCE for shallow region doping density

4. Effect of Intrinsic Region Thickness

The intrinsic region thickness plays a pivotal role in carrier generation and electric field distribution. Fig.8 shows 3D contour plot for i-region thicknesses varying from 0.01 μm to 0.062 μm , both simulated and predicted values of open-circuit voltage (V_{oc}), short- current density, (J_{sc}), fill factor (FF), and power conversion efficiency (PCE) exhibit remarkable consistency. V_{oc} gradually decreases from 1.254 V at 0.01 μm thickness to 1.233 V at 0.062 μm , reflecting the reduction in built-in field with increased thickness. J_{sc} exhibits a steady increase from 30,037 mA/cm^2 to 30,087 mA/cm^2 , indicating that a thicker i-layer absorbs more photons and generates more carriers, but after 0.06 μm J_{sc} starts decreasing because the absorption coefficient drastically falls with the increase of intrinsic region. Fill factor (FF) remains nearly constant at 0.88–0.89, demonstrating minimal loss in device quality due to thickness

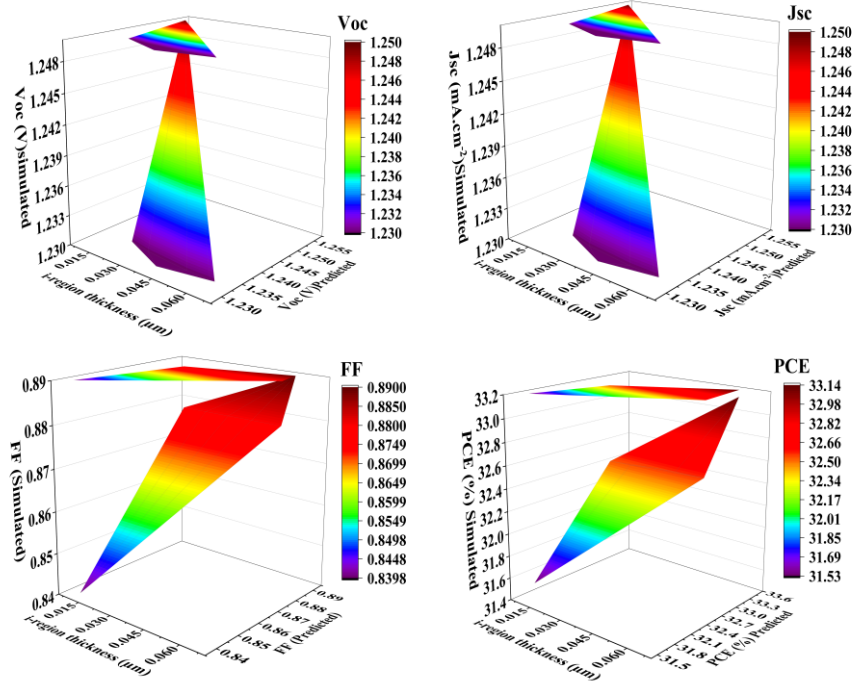


Fig. 8 3D Contour plot of four FOMs for various sets of i-layer width

changes. PCE shows slight improvement from 31.53% to 32.14%, peaking at 0.06 μm , beyond which further improvements become negligible. These results confirm that optimizing the i-region thickness enhances photocurrent without compromising V_{oc} , and the PINN model successfully captures this balance with negligible deviation from simulation results (errors <0.1%). The optimization lies in choosing a thickness that maximizes photon absorption without weakening the field or inducing recombination losses. The PINN model captures these nonlinear trade-offs accurately.

Table: Comparative analysis of the earlier published work with the proposed device outcome

Sl. No.	Device	V _{oc} (V)	J _{sc} (mA.cm ⁻²)	FF	PCE
1.	Device 1 [12]	1.110	8.60	0.790	7.50
2.	Device 2 [13]	0.640	15.20	0.750	7.30
3.	Device 3 [14]	1.180	1.40	0.810	11.85
4.	Device 4 [20]	1.470	4.08	0.790	4.73
5.	Device 5 [22]	0.850	26.45	0.740	16.80
6.	Device 6 [26]	0.780	15.43	0.760	9.14
7.	Device 7 [4]	0.875	27.90	0.815	19.89
8.	This device	1.23	30.27	0.89	33.14

Table: Estimation of errors between simulation data and predicted data for photovoltaics parameters under 1000 sun

PV parameters	Simulated data	Predicted data	Error (%)
V _{oc} (V)	1.23	1.233	0.24
J _{sc} (mA.cm ⁻²)	30276.5	30276.598	<0.003
FF	0.89	0.89	0
PCE	33.14	33.44	0.9

Table 3 presents a comparison of key photovoltaic parameters — Voc, Jsc, fill factor (FF), and power conversion efficiency (PCE) — for our optimized solar cell, which incorporates a 1.4 μm thick lightly doped layer of GaAs with doping concentration of 1018 cm⁻³, and a 0.06 μm thick intrinsic layer, against previously reported devices. The results clearly demonstrate that our optimized solar cell outperforms six previously reported devices [4,12–14, 20, 22, 26] across all four-performance metrics: Voc, Jsc, FF, and PCE, with a PCE of 33.14%.

Additionally, we performed an error analysis by comparing our optimized simulated results with predicted data from PINN model for the optimized device. As shown in Table (), the differences between the two sets of data are less than 1%. This minor deviation is attributed to the cap layer's influence; in simulations, the Auger recombination (AR) model was included, whereas it was omitted in the PINN model. In conclusion, our findings introduce a novel structural and compositional design approach for achieving high-efficiency solar cells.

Conclusion:

This work presents a comprehensive investigation into the photovoltaic performance enhancement of a GaInP/GaAs modified pin solar cell, combining both physics-based simulations and a Physics-Informed Neural Network (PINN) framework. The optimized solar cell design, featuring a lightly doped layer thickness of $1.4\text{ }\mu\text{m}$ with doping concentration of 10^{18} cm^{-3} , and intrinsic layer thickness of $0.06\text{ }\mu\text{m}$ demonstrated a power conversion efficiency (PCE) of 33.14%. Comparative analysis with previously reported devices (Table 3) shows that the proposed design surpasses prior results in all key performance parameters (V_{oc} , J_{sc} , FF, and PCE), demonstrating the effectiveness of carefully engineered layer thicknesses and material composition. Furthermore, error analysis between predicted results and TCAD simulations (Table 4) revealed deviations of less than 1%, confirming the reliability and accuracy of the developed model. Overall, our research findings establish a novel structural and compositional optimization framework for high-efficiency III-V pin solar cells. The successful integration of PINNs, combined with verified simulation models, provides a powerful and computationally efficient tool for rapid solar cell design and performance prediction.

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